



Shyam Banerjee
Bachelors in Statistics
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Course	College/University	Year	CGPA/%
Bachelors in Statistics	Indian Statistical Institute	2024-Present	82.0
Intermediate/+2	Gitanjali Public School	2024	94.8
High School	Gitanjali Public School	2022	97.4

SCHOLASTIC ACHIEVEMENTS

- Secured **rank 9** in Madhava Mathematics Competition [2026]
- Secured **rank 48** in Simon Marais Mathematics Competition [2026]
- Finalist at Trilytics Case Analytics (IIM Calcutta) [2025]
- Cleared IOQM (Regional) [2022]

WORK EXPERIENCE

- Software Engineer Intern | Alumnus Software Ltd.** [Apr'26-Sep'26]
 - PCB Manufacturing Analytics*
 - Built a Streamlit platform for PCB anomaly analysis using a two-stage rule-based and Random Forest classification pipeline.
 - Developed incremental retraining and rule-impact evaluation workflows for manufacturing fault diagnosis.
 - Database Encryption Infrastructure*
 - Implemented transparent database encryption using pg_tde with HashiCorp Vault for secure key management.
 - Configured key storage, retrieval, rotation, and recovery workflows using Vault secret engines.

RESEARCH EXPERIENCE

- Posterior Inference and Gaussian Process Modeling of Stellar Variability | Independent Research** [May'26-Present]
 - Developed a simulation framework for periodic stellar signals corrupted by Gaussian Process noise.
 - Implemented GP-based probabilistic imputation and uncertainty quantification on Kepler stellar light curves.
 - Performed Bayesian posterior inference for hidden periodic structure using harmonic signal models.
 - Investigated inference behaviour across quiet and rotationally variable stars using real Kepler observations.
 - Produced a research manuscript and open-source implementation.

PROJECTS

- Non-stationary Time Series Modeling | Dr. Shyamal Krishna De** [Apr'26-May'26]
 - Developed a seasonal ARIMA forecasting pipeline for weekly PM2.5 concentration data from CPCB Delhi using ~8 years of observations.
 - Performed nonstationarity diagnosis using ADF, KPSS, Mann-Kendall, and Seasonal Mann-Kendall tests, followed by Box-Cox variance stabilisation and iterative seasonal/regular differencing.
 - Implemented forecasting and residual diagnostics in R using `forecast`, `tseries`, and `urca`; evaluated probabilistic forecast coverage on held-out test data.

SKILLS & INTERESTS

- Programming Languages:** Python, C++, R
- Tools & Technologies:** Scikit-learn, Streamlit, PostgreSQL, Vault, Docker, $\text{\LaTeX}$
- Interests:** Time Series, Astrostats, ML, Quantitative Finance

RELEVANT COURSES

- Statistical Inference | Linear Statistical Models | Probability Theory | Stochastic Process | Statistical Methods | Linear Algebra | Design and Analysis of Algorithms | Differential Equations | Sample Survey** [B.Stat]

VOLUNTEERING EXPERIENCE

- Mentoring | AP Tutors** [Sep'25-Mar'26]
 - Guided students for AP Stats and SAT Maths

EXTRACURRICULAR ACTIVITIES

- Competitive Programming (Codeforces)
- Game Theoretic Puzzles